

BELBIN BENO R M

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PROFESSIONAL PROFILE

AI Engineer with a **B.E. in Computer Science (8.53 CGPA)** and a professional background in **Quantitative Analysis** and **Strategic Resource Planning**. Specialized in building **RAG-based Generative AI pipelines**, high-precision **Stacking Ensembles (0.9536 AUC)**, and automated ML utilities. Expert in Python (OOP), FastAPI, and SQL, with a proven ability to modernize industrial workflows into scalable AI solutions. Available for immediate joining.

CORE TECHNICAL SKILLS

- **Generative AI & NLP:** RAG (Retrieval-Augmented Generation), LangChain, Vector Databases (pgvector), LLM API Integration (OpenAI, Gemini, Groq), Prompt Engineering.
- **Machine Learning:** Supervised Learning (XGBoost, LightGBM, CatBoost), Stacking/Voting Ensembles, Hyperparameter Tuning (Optuna), Scikit-Learn.
- **Engineering & Deployment:** FastAPI, LLM, RESTful API Development, Docker (Basics).
- **Data Engineering:** Python (Expert/OOP), SQL (PostgreSQL), ETL Pipelines, Feature Engineering, Automated Data Validation.
- **Tools:** Git/GitHub, Power BI, n8n Automation, Advanced Excel.

PROFESSIONAL EXPERIENCE

Quantitative Analyst | Adroit Design India Pvt Ltd – Chennai

May 2024 – Feb 2026 (1 Year 10 Months)

- **Strategic Resource Planning:** Managed end-to-end project analytics and strategic planning for large-scale infrastructure and construction projects.
- **Cost Estimation & Analytics:** Developed data-driven models for cost estimation, ensuring high-integrity reporting for project budgeting.
- **Workflow Automation:** Transitioned legacy Excel-based civil engineering workflows into automated Python cleaning and analytical scripts, reducing manual processing time by 40%.
- **KPI Reporting:** Audited high-volume data streams to deliver monthly KPI reports that optimized resource allocation and material usage trends.
- **Technical Integration:** Bridged the gap between raw construction data and downstream BI dashboards for executive decision-making.

Machine Learning Practitioner | Kaggle (Competitive Modeling)

2023 – Present

- **Ensemble Optimization:** Consistently applied Stratified K-Fold cross-validation and nested stacking techniques to structured datasets to achieve top-tier AUC scores.
- **Model Explainability:** Developed reproducible Jupyter Notebooks focusing on feature selection, importance gain, and model interpretability (SHAP/LIME).
- **Feature Engineering:** Specialized in handling high-kurtosis data and extreme outliers using advanced clipping and transformation strategies.

KEY PROJECTS

AIMO: Multi-LLM Report Stacking & API Deployment

- **Objective:** Build a high-fidelity reporting engine by stacking outputs from multiple LLMs (GPT, Llama, Qwen) to reduce "hallucinations" and ensure factual consistency.
- **Architecture:** Developed a **Stacking Logic** for text-similar to your 0.9536 AUC tabular stack-where a "Meta-Evaluator" LLM synthesizes the best insights from base model reports.
- **Deployment:** Served the entire pipeline via **FastAPI**, creating a RESTful endpoint that handles asynchronous requests for real-time report generation.
- **Impact:** Integrated **vLLM** for inference acceleration, ensuring the API could handle high-volume document requests with minimal latency.

Precision-5 Diagnostic Stacking Architecture (Advanced ML)

- **Objective:** Optimize prediction accuracy using a multi-model approach for complex diagnostic datasets.
- **Architecture:** Engineered a 5-model Stacking Classifier (XGBoost, LGBM, CatBoost, HGB, and Logistic Regression) with a RidgeCV meta-learner.
- **Innovation:** Built Custom Scikit-Learn Transformers (OOP) to isolate model-specific features and handle interaction terms, achieving a peak **0.9536 AUC**.
- **Result:** Demonstrated high model stability with minimal variance between local validation and leaderboard scores through rigorous OOF (Out-of-Fold) strategies.

Document Intelligence & Knowledge Retrieval (RAG/GenAI)

- **Objective:** Build a scalable pipeline to extract insights from unstructured PDF reports and technical documentation.
- **Implementation:** Developed a Retrieval-Augmented Generation (RAG) system using **LangChain** and **pgvector** for semantic similarity search.
- **Impact:** Reduced document review time by 40% by implementing grounded summaries and intelligent chat workflows for data-intensive files.
- **Tech Stack:** Python, OpenAI API, PostgreSQL, Vector Embeddings, FastAPI.

Operational Forecasting & ML Analytics Platform

- **Objective:** Automated material usage forecasting using time-series analysis for industrial resource management.
- **Execution:** Engineered time-based features and implemented batch prediction pipelines using Regression and Prophet models.
- **Result:** Enabled proactive resource allocation, directly reducing project waste and overhead costs.

EDUCATION

- **B.E. Computer Science Engineering** | Anna University | 2024 | **CGPA: 8.53**
- **PGP – Data Analytics & Data Science** | Skill-Lync | 2024

CERTIFICATIONS

- Advanced Machine Learning & Deep Learning – Skill-Lync
- SQL for Data Science – Skill-Lync
- Core & Advanced Python Programming – Skill-Lync